

LM-Lexicon: Improving Definition Modeling via Harmonizing Semantic Experts

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Abstract

LM-Lexicon is a definition modeling approach using data clustering, semantic expert learning, and sparse mixture-of-experts architecture. By dividing the task into specialized semantic domains with small language models as experts, LM-Lexicon achieves substantial improvements (+7% BLEU score) over existing methods on five benchmarks. We show that 1) clustering enables fine-grained specialization with nearly 10% improvement; 2) semantic-aware domain-level routing achieves higher efficacy (+1%) than token-level routing; and 3) further gains come from test-time compute and semantic expert scaling. Our work advances definition modeling while providing insights into efficient language models for semantic-intensive applications.



1. The Challenge: Semantic Heterogeneity

Current DM models struggle with:

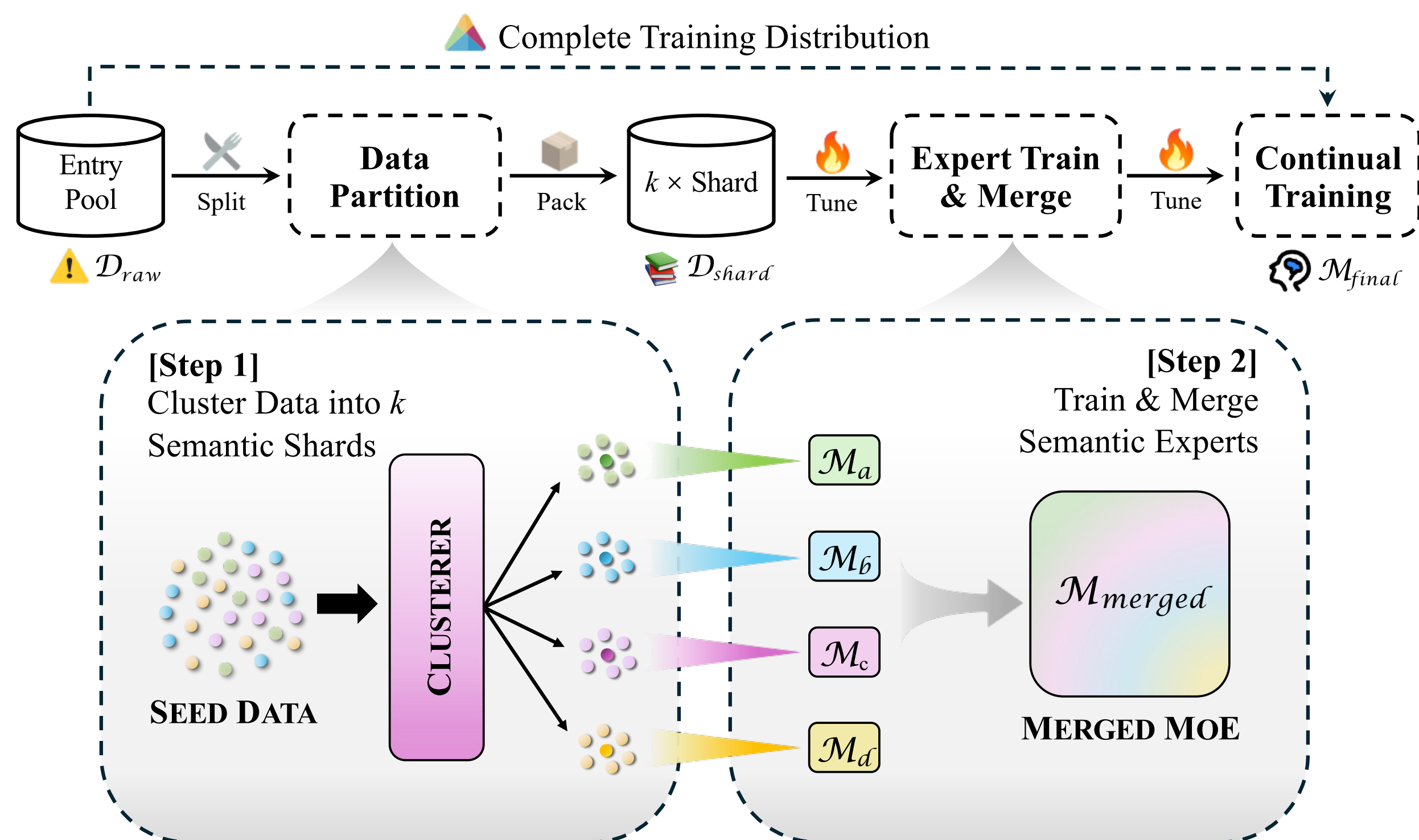
- **Subtle Word Senses:** Difficulty capturing rare or domain-specific meanings.
- **Superposition:** Dense architectures force distinct meanings to share parameters.
- **Generalization:** Frontier LLMs often produce definitions that are either too generic or too specific.

2. Methodology: Specialize-then-Synthesize

Step 1. Data Partitioning: We cluster training data into $N = 4$ semantic shards (e.g., Scientific, Adjectives, Person Names, Proper Nouns) using balanced k-means on semantic embeddings.

Step 2. Semantic Experts: We train N small language models as "Domain Experts" specialized in specific semantic clusters.

Step 3. MoE Merging: Experts are merged into a single MoE model. The Feed-Forward Networks (FFNs) become expert layers, while remaining parameters are averaged. We use **domain-level sequence routing**.



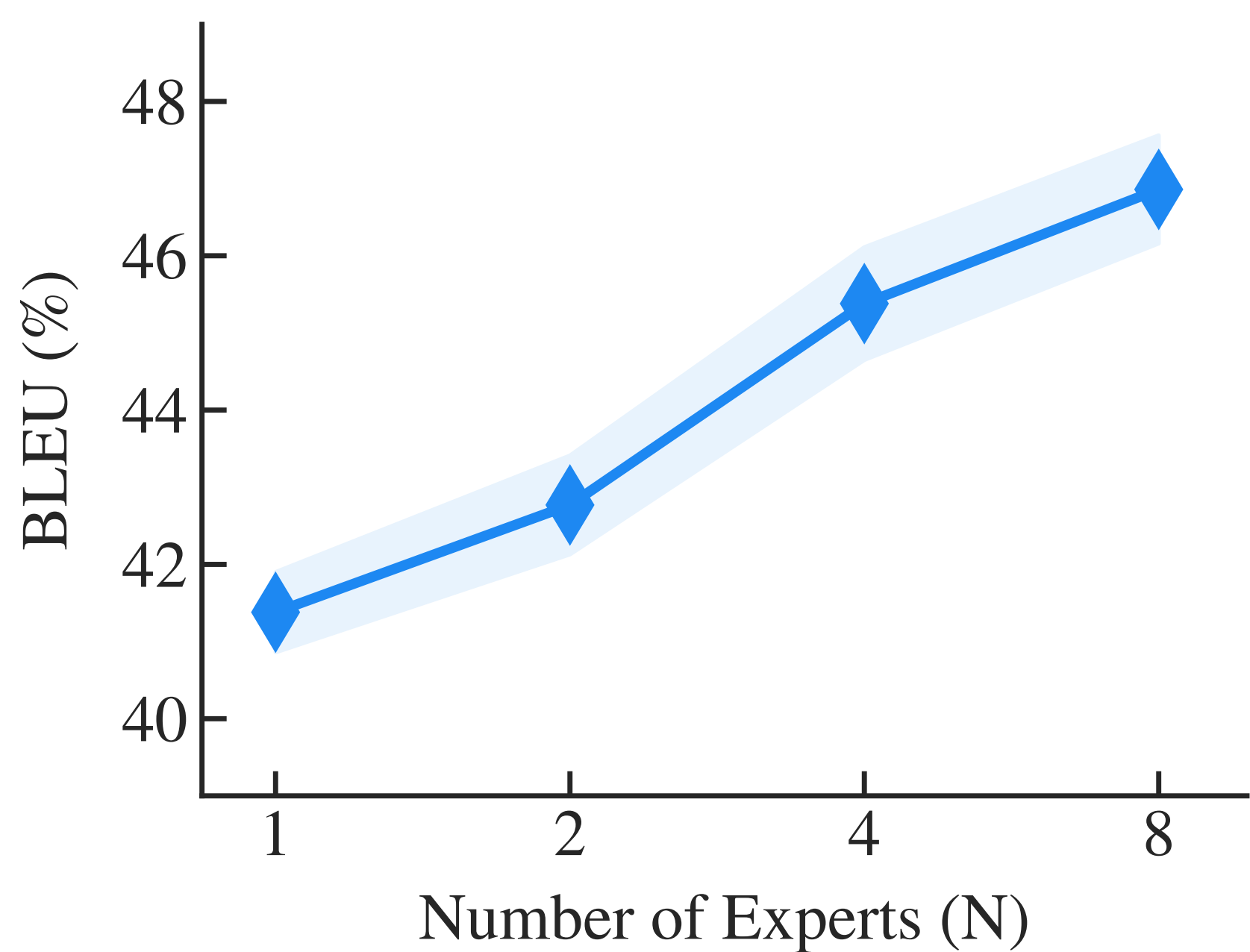
3. Domain-Level Routing

Unlike token-level routing, our **domain-level policy** routes the entire sequence to the nearest semantic centroid.

$$\text{FFN}_{\text{MoE}}^{\ell}(x) = \sum_{i=1}^N \mathcal{G}(x) \cdot \text{FFN}_i^{\ell}(x)$$

This ensures that semantic-intensive tasks benefit from domain-specific specialization.

Scaling Semantic Experts



Algorithm 1 Compose MHA and MLP modules for each decoder layer ℓ in LM-LEXICON.

Input: Domain Experts $\mathcal{E} := \{e_1, e_2, \dots, e_n\}$.
Output: LM-LEXICON-MoE ($\mathcal{M}$)

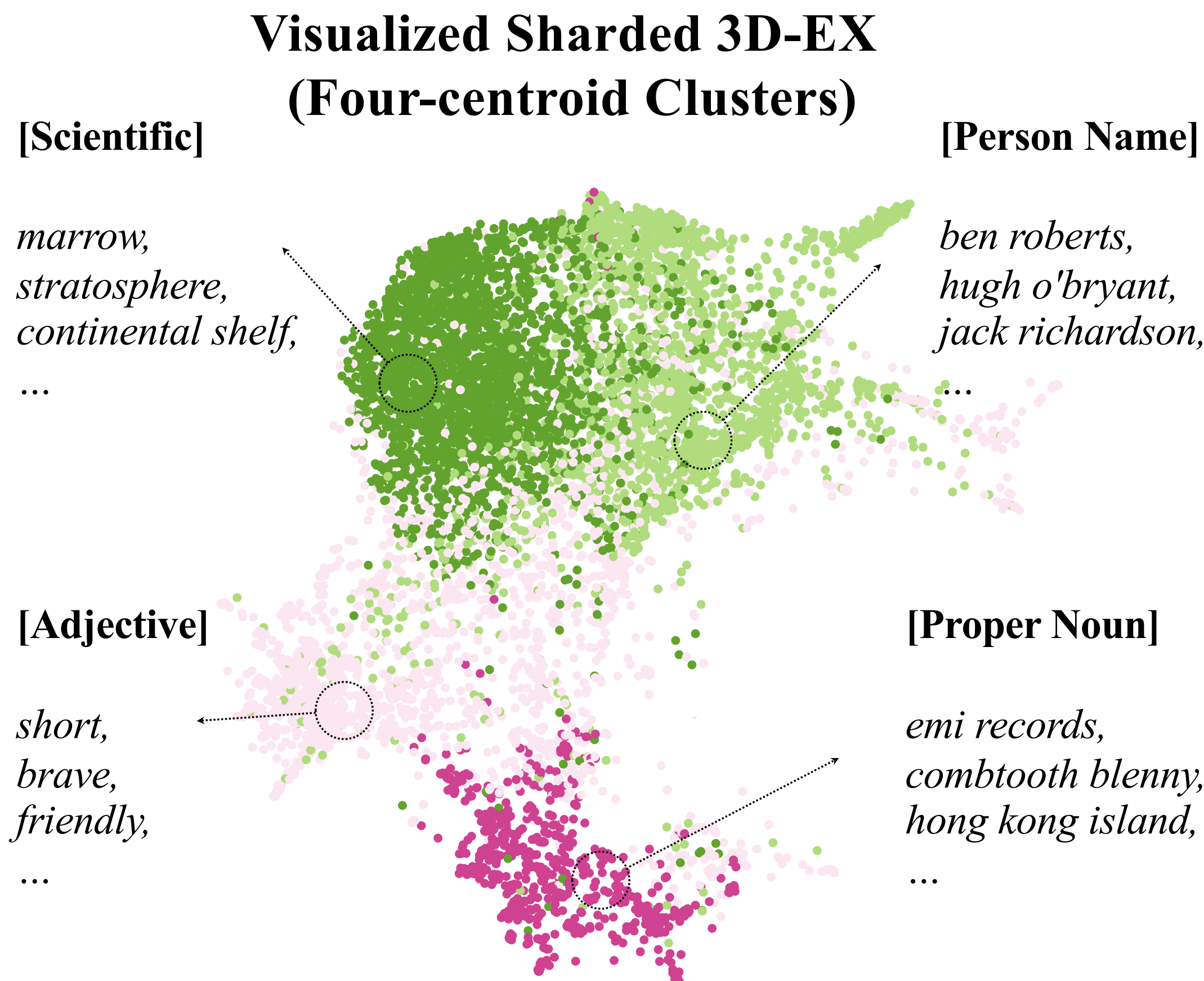
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1: procedure MODULES-COMPOSER( $\mathcal{E}$ )
2:    $\mathcal{M} \leftarrow \emptyset$  ▷ INIT STATE DICT
3:   for  $e_i \in \mathcal{E}$  do ▷ ITERATE EACH EXPERT
4:      $i \leftarrow \text{GetExpertIdx}(e_i)$ 
5:     /* Retrieve MHA and MLP weights */
6:      $\theta_{mha}, \theta_{mlp} \leftarrow \text{HookWeights}(e_i)$ 
7:     for  $\theta \in \{\theta_{mha}, \theta_{mlp}\}$  do
8:       if IsRouterLayer( $\theta$ ) then
9:         /* Get formatted layer name */
10:         $n \leftarrow \text{FormatName}(\theta, i)$ 
11:         $\mathcal{M}[n] \leftarrow \theta$ 
12:       else ▷ AVERAGE  $\theta$  OF MODULE
13:         $\mathcal{M}[n] \leftarrow \mathcal{M}.get(n, 0) + \theta / |\mathcal{E}|$ 
14:   return  $\mathcal{M}$ 

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4. Visualizing the Lexical Semantic Space

UMAP visualization of 10K random definitions in the 3D-EX dataset shows clear separation between the four clusters.



5. Experimental Results

Table 1 shows LM-Lexicon outperforms baselines and SoTA methods. Key findings: (1) 10% BLEU/ROUGE gains on 3D-EX over prior SoTA. (2) Best scores on Urban dataset (31.26% BLEU, 33.81% ROUGE), showing efficacy for rare word senses. (3) Surpasses frontier LMs by 23.44%, 9.14% on Wiki, WordNet BLEU. (4) Lower Oxford performance due to short terms with long contexts and 80% term overlap with near-zero term-definition overlap, indicating strong polysemy limiting generalization. Overall, it confirms effectiveness of semantic expert specialization and semantic-focused sparsifying upcycling.

Model	WordNet	Oxford	Wiki	Urban	3D-EX	Avg.
Rerank-T5	30.9	28.0	57.2	18.2	38.5	34.6
GPT-4-Turbo	29.7	34.3	40.6	26.5	37.6	33.7
Claude-3-Opus	21.7	25.9	15.8	7.1	24.6	19.0
LM-Lexicon (4×8B)	40.6	33.7	57.6	33.8	46.3	42.3

Table 1: Performance comparison (Rouge-L). LM-Lexicon achieves ~7-10% improvement.

6. Human Evaluation

Three linguistics graduates rated models on a 1-5 scale. LM-LEXICON-MoE achieved the highest scores in **Accuracy (4.6)** and **Context Appropriateness**. It consistently outperforms other models in most dimensions, with particularly strong performance of accuracy. While all models demonstrate competent performance with scores above 3.8, LM-Lexicon-MoE shows notable advantages in capturing contextual nuances and maintaining clarity and conciseness in definitions.

